

Week 10: Basic Operations on Whole Numbers

JSS1 Mathematics — First Term Lesson Note (Deep-Dive Edition)

Introduction

Addition, subtraction, multiplication, and division are the four basic operations used constantly in daily life — from calculating change in the market to working out how long a journey will take. This week reviews all four, plus the order in which to apply them.

Addition

Add digits column by column, starting from the units, carrying over when a column totals 10 or more.

Example 1: $3,458 + 2,769$

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3458
+ 2769
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6227
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Example 2: $15,672 + 8,349 = 24,021$

Subtraction

Subtract column by column, starting from the units, borrowing from the next column when needed.

Example 1: $7,203 - 2,846 = 4,357$

Example 2: $10,050 - 4,876 = 5,174$

Multiplication

Multiply digit by digit, then add the partial products.

Example 1: 234×12

$234 \times 2 = 468$

$234 \times 10 = 2,340$

Total: $468 + 2,340 = 2,808$

Example 2: 156×23

$156 \times 3 = 468$

$156 \times 20 = 3,120$

Total: $468 + 3,120 = 3,588$

Division

Division shares a number into equal groups. It can leave a **remainder**.

Example 1: $156 \div 7 = 22$ remainder 2 (since $7 \times 22 = 154$, and $156 - 154 = 2$)

Example 2: $245 \div 5 = 49$ (since $5 \times 49 = 245$, no remainder)

Order of Operations (BODMAS)

When a problem has more than one operation, follow **BODMAS**: **B**rackets → **O**rders (powers/roots) → **D**ivision/**M**ultiplication → **A**ddition/**S**ubtraction

Example 1: $4 + 3 \times (6 - 2) = 4 + 3 \times 4 = 4 + 12 = 16$

Example 2: $(10 - 4) \times 2 + 5 = 6 \times 2 + 5 = 12 + 5 = 17$

Real-Life Application

A trader buys 24 crates of soft drinks at ₦2,808 per crate (from the multiplication example above). To find the total cost, she multiplies; to work out change after paying with a larger note, she subtracts; and to share a total profit equally among 7 business partners, she divides.

Vocabulary

Word	Meaning
Sum	The result of addition
Difference	The result of subtraction
Product	The result of multiplication
Quotient	The result of division

Evaluation Questions

- 1 Add: $5,672 + 3,849$
- 2 Subtract: $9,004 - 3,567$
- 3 Multiply: 345×24
- 4 Divide: $288 \div 9$
- 5 Use BODMAS to solve: $20 - (3 \times 4) + 2$
- 6 Add: $12,345 + 6,789$
- 7 Multiply: 128×15
- 8 Use BODMAS to solve: $(8 + 2) \times 3 - 5$
- 9 Divide 315 by 9, stating the quotient and any remainder.
- 10 Explain, using your own example, why BODMAS matters when a problem has more than one operation.

(Weeks 5, 7, and 11-13 are Midterm Exams, Midterm Break, and Revision/Exams respectively — no new content covered.)